

Manual

Quality Data Analysis Suppliers/Goods Receipt/Production SMA-QDA 8.1

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1 Quality Data Analysis Suppliers/Goods Receipt/Production

Purpose

The *Quality data analysis* contains the following SMA applications.

- Failure mode analysis
- Analysis of usage decisions
- Analysis of inspection quality and costs
- Comparison of assessed parties
- Development of an assessed party
- Criterion analysis

The quality data analysis is based on the performed supplier evaluations and the inspection data from goods receipt and production. The inspection data is made up of the inspection points including detail information, failures recorded in the inspection process and data of the inspection requirements.

All applications provide graphic evaluations using pivot analyses. In addition to the graphic evaluations, the filtered data can also be displayed in tables.

You can define different application profiles for each application. This way, you can predefine individualized evaluation profiles for specific use cases. You can change the evaluation profile manually or you can select another application profile. Another option is to manually change the evaluation profile.

Implementation notes

To evaluate the product quality and to identify methods to improve quality, the applications of the *Quality data analysis* in combination with the actions taken are the ideal products.

Integration

You can integrate the applications of the *Quality data analysis* in the application *Information Dashboard*.

Features

The following functions are available:

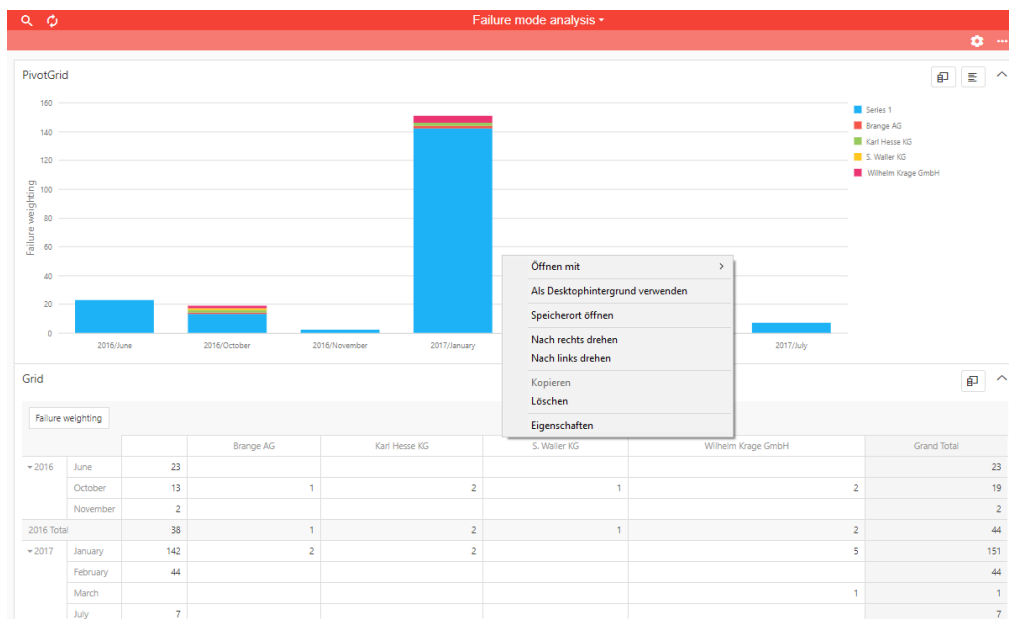
- Graphic and tabular analysis with extensive filter criteria and display options.
- Failure mode analysis to evaluate recorded failure types, locations and causes for each characteristic, article, machine, supplier, month, etc.
- Display of the development of inspection costs per day, month, quarter, year
- Calculation of inspection costs for each article, article group, machine group, operation, tool.
- Analysis of inspection point decisions

- Monitoring of the development of the inspection reliability in a day, week, month, quarter, year (ratio between completed and open inspection points to the total number of inspection points).
- Analysis of the goods received regarding:
 - usage decisions (release, special permit, rework, lock, etc.)
 - processing times and their development in a month, quarter and year.
- Monitoring of supplier evaluations:
 - Analysis and development of evaluation criteria (progress) in the course of evaluations performed at cyclic intervals.
 - Development of evaluated suppliers
 - Comparison of suppliers

2 Failure mode analysis

Overview

App name	Failure mode analysis
Short name of app	Failure mode analysis
Function authorization	SMA.FMA



Use the *Failure Mode Analysis* to evaluate failures of the following types in a pivot analysis. These failures are collected during the inspection process or generated automatically.

- Failure type
- Failure location
- Failure cause

In addition to the graphic view, you can also display the data in a table.

Purpose

The application *Failure mode analysis* includes the following features:

- Graphic and tabular analysis of the failures recorded manually or generated automatically during the inspection process.
- Monthly presentation of the failure development.
- Identification of the failure frequency per article, failure group, machine, machine group, operation, tool, etc.
- Analysis of variations in quality to initiate preventive and corrective action with the aim to avoid failures in future.

Integration

The *Failure mode analysis* evaluates failures of inspection requirements from the following area types:

- Goods receipt
- Initial sample inspection
- Production
- Goods issue

Consequently, the *Failure mode analysis* is a general analysis tool.

Requirements

Requirements: You have to collect the failures manually or the system must generate the failures automatically. You have to define the failures beforehand in the master data of Quality Management.

Selection criteria

Use the button  to open the selection area.

Use the configuration button  of the selection area to open the fields that are available for this application. Select the required fields to enable these fields as filter fields.

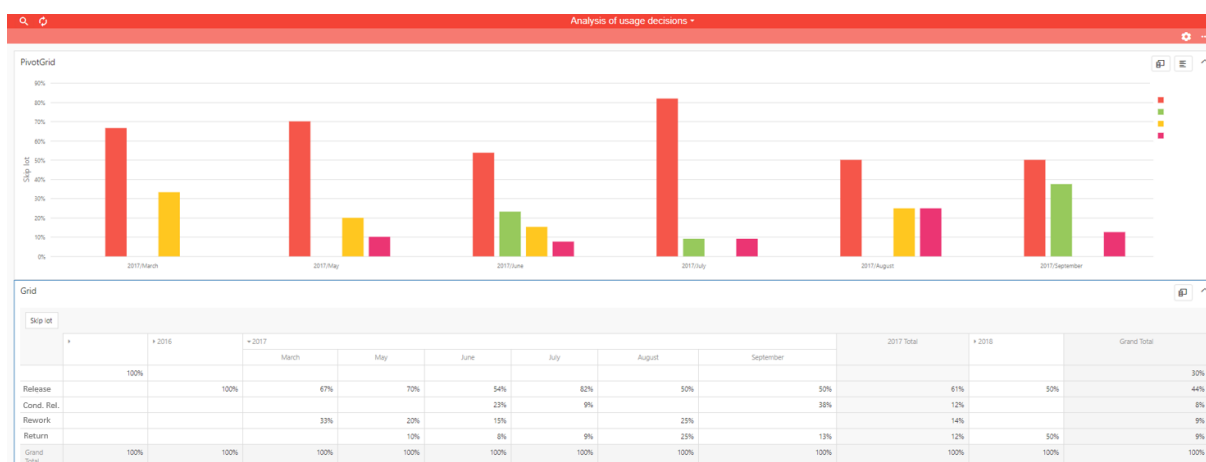
Editing functions

This application does not include any editing functions.

3 Analysis of usage decisions

Overview

App name	UsageDecisionDurationAnalysis
Short name of app	UsageDecisionDurationAnalysis
Function authorization	SMA.UDDA



Use the application *Analysis of usage decisions* to analyze the usage decisions of inspection requirements in a graphical pivot analysis. In addition to the graphic view, you can also display the data in a table.

Purpose

The application *Analysis of usage decisions* mainly evaluates goods receipt inspection requirements.

The application *Analysis of usage decisions* includes the following functions:

- Graphical and tabular analysis of inspection costs based on the inspection duration of an inspection requirement.
The inspection duration results from the period between creation and completion of the respective inspection requirement.
- Display of the development of inspection costs per day, month, quarter, year.
- Graphical and tabular analysis of usage decisions of inspection requirements. Possible usage decisions are:
 - Release
 - Special permit
 - Rework

- Rejection
- Analysis of the development of usage decisions per month, quarter, year.

Integration

The application *Analysis of usage decisions* is directly connected with the *Inspection requirements* application.

Once you have completed an inspection requirement, the system identifies the time difference between creation and completion of the inspection requirement and stores this time as the inspection duration for the inspection requirement. The inspection duration is the basis for analyzing inspection costs.

The usage decision resulting from the completion of an inspection requirement is the basis for analyzing the development of usage decisions.

The application *Inspection requirements* shows the inspection duration of an inspection requirement and its usage decision. In the application *Inspection requirements*, you can complete inspection requirements with any usage decision you want if you have the right permissions. If the system completes inspection requirements automatically, the usage decision is always based on the inspection results.

Requirements

You have to use the in-production inspection or the goods receipt inspection in order to complete inspection requirements.

Selection criteria

Use the button  to open the selection area.

Use the configuration button  of the selection area to open the fields that area available for this application. Select the required fields once more to enable these fields as filter fields.

Editing functions

This application does not provide editing functions.

4 Analysis of Inspection Quality and Costs

Overview

App name	Inspection cost and inspection quality
Short name of app	Inspection cost and inspection quality
Function authorization	SMA.IQEA



You can display the data in a the graphic view and in a table.

Purpose

The application *Analysis of inspection quality and costs* provides the following functions:

- Graphic and tabular analysis of inspection costs using the time required for the inspection of one inspection point.
The time required for the inspection is the time between the first and the last measured value recorded for an inspection point.
- Display of the development of inspection costs per day, month, quarter, year.
- Identification of the inspection costs per article, article group, machine group, operation, tool, etc.
- Graphic and tabular analysis of inspection point decisions.
- Visualization of the development of the inspection reliability per day, week, month, quarter, year.
The inspection reliability is the ratio between completed and open inspection points to the total number of inspection points.

Integration

The application *Analysis of inspection quality and costs* is directly connected to the functions to collect inspection data using AIP and SMA and the application *Inspection points*.

When you record inspection data, the measured values and attributive inspections are recorded for the characteristics of an inspection point. The difference between the first and the last inspection for an inspection point characteristic is called inspection time or duration and is stored as information on the inspection point. To analyze inspection costs, the inspection duration is used.

When you complete an inspection point, you take a usage decision. This usage decision is the basis for the analysis of the inspection quality.

The application *Inspection points* shows the inspection duration and the usage decision. If you are an authorized user, you can also change the inspection point decision in the application *Inspection points* and/or complete the inspection point.

Requirements

You must record data for an inspection point. And the inspection point must be completed so that the inspection duration can be calculated and an inspection point decision is available.

Selection criteria

Use the button  to open the selection panel.

Use the configuration button  in the selection panel to open the fields that are available for this application. Select the required fields to enable these fields as filter fields.

Editing functions

This application does not provide any editing functions.

5 Comparison of Assessed Parties

Overview

App name	Comparison of assessed parties
Short name of app	Comparison of assessed parties
Function authorization	SMA.AOC



The application *Comparison of assessed parties* provides a graphic comparison of evaluation results of different assessed parties, e.g. suppliers. The comparison is provided as pivot chart analysis. The comparison of assessed parties is based on the overall results of supplier evaluations that were made during a specified period of time.

In addition to the graphic view, you can also display the data in a table.

Purpose

The application *Comparison of assessed parties* includes the following functions:

- Evaluation results of different assessed parties, e.g. suppliers, are compared in graphic and tabular form. The comparison is based on the overall results of supplier evaluations performed.
- The classification class of assessed parties is displayed.
- The development of the overall results of performed supplier evaluations is displayed and compared for different evaluation periods.

Integration

This evaluation only uses assessments from the *Assessment management*.

Requirements

There are no special requirements. This application only requires supplier evaluations, which must have been made for the *Assessment management*.

Selection criteria

Use the button  to open the selection panel.

Use the configuration button  of the selection panel to open the fields that are available for this application. Use a separate identifier to enable these fields as filter fields.

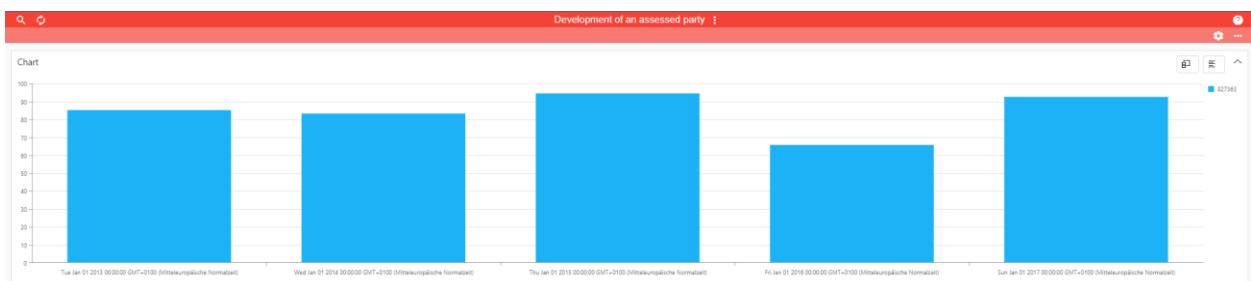
Editing functions

This application does not include any editing functions.

6 Development of an Assessed Party

Overview

App name	Development of an assessed party
Short name of app	Development of an assessed party
Function authorization	SMA.AOH



The application *Development of an assessed party* provides a graphic analysis of an assessed party.

Example: You can view the overall result of supplier assessments over a specified period of time in form of a pivot analysis.

In addition to the graphic view, you can also display the data in a table.

Purpose

The application *Development of assessed parties* includes the following functions:

- Graphic and tabular analysis of the overall assessments based on the supplier evaluations made.
- The development of the overall results for performed supplier evaluations is displayed.
- The resulting classification (class) of a supplier/assessed party is visualized for each supplier evaluation made.

Integration

This evaluation only uses assessments from the *Assessment management*.

Requirements

There are no special requirements. This application only requires supplier evaluations, which must have been made for the *Assessment management*.

Selection criteria

Use the button  to open the selection panel.

Use the configuration button  of the selection panel to open the fields that are available for this application. Use a separate identifier to enable these fields as filter fields.

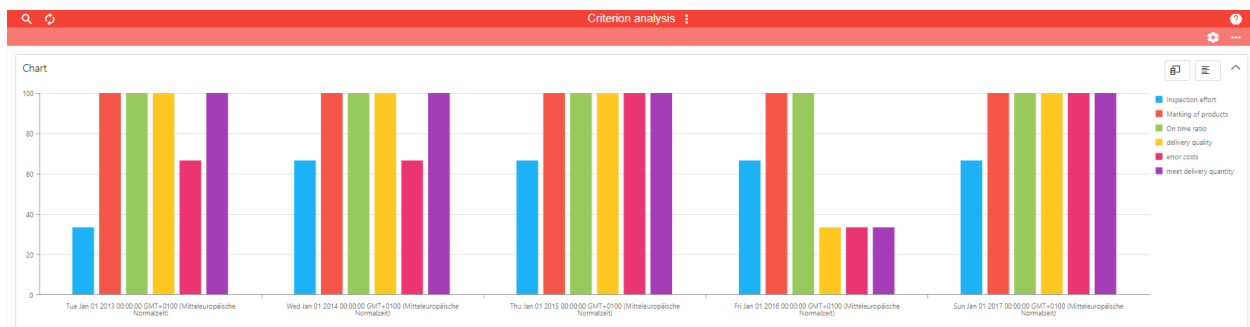
Editing functions

This application does not include any editing functions.

7 Criterion Analysis

Overview

App name	Criterion analysis
Short name of app	Criterion analysis
Function authorization	SMA.EACA



The application *Criterion analysis* provides a graphic analysis of the development of assessment criteria from supplier evaluations during a specified period of time. The analysis is provided in form of a pivot analysis. You can also compare the different assessment criteria in a specified analysis period.

In addition to the graphic view, you can also display the data in a table.

Purpose

The application *Criterion analysis* includes the following functions:

- Graphic and tabular analysis of the assessment criteria based on the supplier evaluations made.
- Comparison of the classification of assessment criteria when a supplier evaluation was made.
- Display of the development of how assessment criteria were classified in a specified period of time where several supplier evaluations were made.

Integration

This evaluation only uses assessments from the *Assessment management*.

Requirements

There are no special requirements. This application only requires supplier evaluations, which must have been made for the *Assessment management*.

Selection criteria

Use the button  to open the selection panel.

Use the configuration button  of the selection panel to open the fields that are available for this application. Use a separate identifier to enable these fields as filter fields.

Editing functions

This application does not include any editing functions.